The Euclid–Kepler–Pacioli Cell

A periodic cell in which all five Platonic solids nest on 13 points, and a corner-sharing network of Kepler stars and icosahedra

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Version 1 · 15 October 2026 · DICTO DOI: 10.5281/zenodo.23284106 · Licence CC BY 4.0 · Data, renders and checks are included with this record. First released inside Kaleidohedra v2026.10.11-octet (DICTO DOI: 10.5281/zenodo.23284390); this is its own record.
The construction, its naming and the network are DICTO's. Computation and checking with Claude (Anthropic) in the open-source library krp-core (github.com/DICTOR-Master/krp-core, v0.16.0).

Abstract. Place Euclid's six roofs on a cube of edge 2 to obtain the regular dodecahedron, then reflect the twelve roof vertices back through the cube faces: they land exactly on a regular icosahedron inside the cube. Repeating the cube by translations of 2 gives a periodic structure, which we call the **Euclid–Kepler–Pacioli (EKP) cell**, with 13 nodes per cell, edge lengths in the chain $2 : 2/\varphi : 2/\varphi^2$ (cube : dodecahedron : icosahedron), and space group Pm-3 (No. 200), nodes on Wyckoff positions 1b and 12j $(0, y, z)$ with $y = 1/(2\varphi)$, $z = 1/(2\varphi^2)$. Each roof vertex is a vertex of a neighbouring cell's icosahedron, and the ridges of the six neighbouring cells meet inside every cube as Pacioli's three interlocking golden rectangles. The cell nests all five Platonic solids (icosahedron ⊂ octahedron ⊂ stella octangula ⊂ cube ⊂ dodecahedron) with Kepler's great stellated dodecahedron around the icosahedron. From the same node set, great stellated dodecahedra on the even cells and icosahedra on the odd cells share only corners: the **EKP network**. All statements are checked by 48 exact or sampled tests. The individual nestings are classical; the periodic cell as a whole and the EKP network were not found in a literature search (2026-10-06, extended 2026-10-15) and we present them as candidates, not as priority claims.

1. Construction

Take the cube with corners $(\pm1, \pm1, \pm1)$ and $\varphi = (1+\sqrt{5})/2$. Euclid's roof construction (*Elements* XIII.17) adds over each face a roof whose ridge has length $2/\varphi$ and whose ridge vertices are the twelve points $(0, \pm1/\varphi, \pm\varphi)$ and their cyclic permutations; with the cube's eight corners they are the 20 vertices of a regular dodecahedron of edge $2/\varphi$. Reflecting each ridge vertex back through the cube face it stands on, $(0, \pm1/\varphi, \pm\varphi) \to (0, \pm1/\varphi, \pm(2-\varphi)) = (0, \pm1/\varphi, \pm1/\varphi^2)$, gives twelve points that are the vertices of a regular icosahedron of edge $2/\varphi^2$, centred in the cube. The cube is then repeated by translations of 2 along x, y and z (Fig. 4).

The reflection is the fold of the title: on every axis the coordinate is folded back at the cube face, so a roof vertex of one cell *is* an icosahedron vertex of the neighbouring cell. This is why the lattice has only 13 nodes per cell (one cube corner and the 12 icosahedron vertices) and why every dodecahedron vertex is already a node.

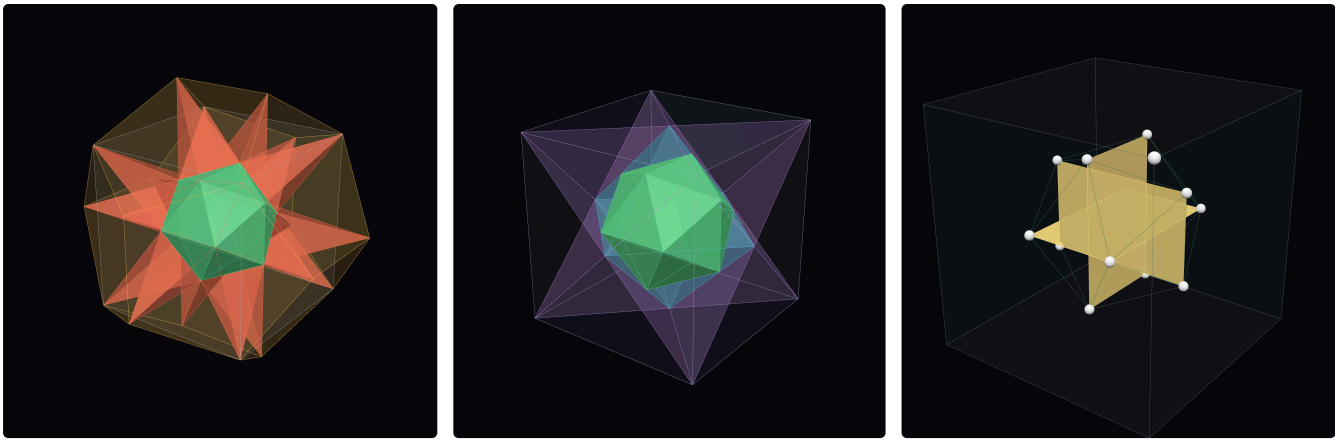


Fig. 1–3. (1) One cell: the cube (grey), Euclid's dodecahedron (gold), Kepler's great stellated dodecahedron (orange) and the icosahedron (green), all on the same 20 + 12 points. (2) Kepler's chain: icosahedron ⊂ octahedron (cyan) ⊂ stella octangula (violet) ⊂ cube. (3) Pacioli's three golden rectangles ($2/\varphi \times 2/\varphi^2$) whose 12 corners are the icosahedron; each is a pair of ridges from the neighbouring cells.

2. Measured properties

Property	Value
Edges	cube 2, dodecahedron $2/\varphi$, icosahedron $2/\varphi^2$ (ratio $\varphi^2 : \varphi : 1$); every edge along one of the icosahedron's 15 two-fold axes
Nodes per cell	13: one cube corner + 12 icosahedron vertices
Space group	Pm-3 (No. 200), symmorphic; 1b (cube corner) and 12j (0, y, z), $y = 1/(2\varphi) \approx 0.3090$, $z = 1/(2\varphi^2) \approx 0.1910$
Aligned axes	each cell's solids carry the icosahedron's 31 rotation axes (6 five-fold, 10 three-fold, 15 two-fold), parallel in every cell; 7 of them are axes of the crystal (3 two-fold, 4 three-fold). m-3 is exactly the set of cube symmetries that are also icosahedron symmetries.
Spikes	over each of the 20 icosahedron faces, exactly one node at distance $2/\varphi$ from its three corners; spike faces are golden triangles (apex 36°); the 20 tips are the dodecahedron's vertices
Great stellated dodecahedron	the icosahedron with its 20 spikes: 60 triangles in 12 planes forming 12 regular pentagrams, volume ≈ 2.918; its convex hull is the dodecahedron
Coverage by dodecahedra	not a tiling: space is covered once (19.1%) or twice (80.9%), mean $(5+\sqrt{5})/4 \approx 1.809 = (10+2\sqrt{5})/8$
Neighbours	face neighbours' dodecahedra overlap by exactly two roofs; edge neighbours touch only; corner neighbours share one vertex
Colourings	x+y+z parity → Fm-3 (No. 202); x+y → Cmmm (No. 65); z → Pmmm (No. 47); eight octants → Pmmm on a doubled cell

3. Pacioli's rectangles are the neighbours' roofs

Each roof ridge pokes into the neighbouring cell. Seen from inside that cell, the ridge is the long side ($2/\varphi$) of a golden rectangle whose short side ($2/\varphi^2$) is an icosahedron edge. The six neighbours' ridges make, inside every cube, the three mutually perpendicular, interlocking golden rectangles of Pacioli's *De divina proportione* (1509), whose twelve corners are the icosahedron (Fig. 3). The fold is therefore literally the neighbours' roofs meeting in the middle of the cell. (Our observation, 2026-10-06: "four elements of roofing structure overlap, making golden section rectangles".)

4. All five Platonic solids in one cell

The two regular tetrahedra on alternate cube corners (Kepler's stella octangula) intersect in the octahedron on the six cube-face centres. The twelve icosahedron vertices lie one on each

of that octahedron's twelve edges, dividing it in the golden ratio, and eight of the icosahedron's faces lie in the octahedron's eight face planes. Hence one cell contains, exactly nested, **icosahedron ⊂ octahedron ⊂ tetrahedra ⊂ cube ⊂ dodecahedron**, with the great stellated dodecahedron between the icosahedron and the dodecahedron (Fig. 1–2). Each inclusion is classical; what is new here is that the whole chain is one periodic cell. The tetrahedra and octahedron edges lie along cube face diagonals (√2, 2√2), not along icosahedral axes, so a physical model needs a second family of rods.

5. Extractions without overlap; the EKP network

The overlap belongs to the solids chosen, not to the points. From the same 13-node set:

(a) Dodecahedra on the even cells only (an FCC lattice) touch their twelve nearest neighbours and never overlap; the density is exactly $(5+\sqrt{5})/8 \approx 0.9045$. This is the known optimal lattice packing of the regular dodecahedron (Betke & Henk 2000; Torquato & Jiao 2009), recovered here inside the structure. **Status: known.**

(b) Great stellated dodecahedra on the even cells (touching tip to tip at shared cube corners) together with icosahedra on the odd cells share *only corners*: each star's twelve roof tips are vertices of the six neighbouring icosahedra, and an exact separating-axis test over 294 piece pairs finds no overlap (Fig. 6). We call this corner-sharing arrangement of Kepler stars and icosahedra the **Euclid–Kepler–Pacioli network**. **Status: not found in a literature search; candidate.**

Dodecahedra on the even cells with icosahedra on the odd cells do overlap; it is the stars that make (b) corner-sharing.

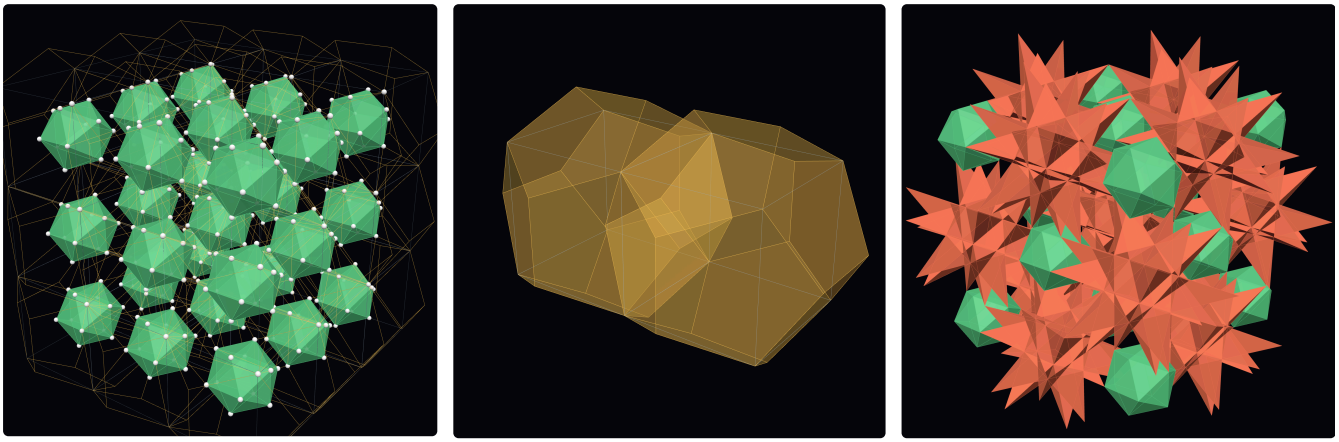


Fig. 4–6. (4) A 3 × 3 × 3 block of cells: icosahedra, dodecahedron edges (gold), cube edges and the 13 nodes per cell (white). (5) Two face neighbours: their dodecahedra overlap by exactly two roofs. (6) The EKP network: great stellated dodecahedra on even cells, icosahedra on odd cells, meeting at corners only.

6. Relation to known work, and status

Classical: Euclid's roof construction of the dodecahedron on a cube; the pyritohedrally placed icosahedron in a cube; the stella octangula, the octahedron and the golden-section points on its edges; Pacioli's three golden rectangles; Kepler's great stellated dodecahedron. Periodic crystals with icosahedral or dodecahedral motifs are well known and are of the same symmetry kind (no five-fold crystal axis): α-AlMnSi (Pm-3, the 1/1 cubic approximant, Mackay icosahedra), skutterudites (Im-3, icosahedral cages on 24g (0, y, z)) and type-I clathrates (Pm-3n). Closest of all are the Tsai-type 1/1 approximants: in AuGeYb (Deguchi et al. 2015) and

ZnAuYb (Ishimasa 2019) the cluster's third shell is an icosahedron of Yb or Tb atoms on 24g at (0, 0.187, 0.305) and (0, 0.188, 0.301), nearly the EKP cell's 12j point (0, 0.309, 0.191) with y and z exchanged, which is the same icosahedron in the other choice of axes. It is about 2% smaller and slightly irregular (ratios 0.613 and 0.626 against $1/\varphi = 0.618$), in a body-centred cell with other shells around it; the roofs, the fold onto the neighbours and the network are not part of those structures. The only packing study of great stellated dodecahedra found (de Graaf, van Roij & Dijkstra 2011) gives a two-orientation dimer lattice with interlocking spikes and no icosahedra, which differs from (b). Construction-kit and mathematical-art sources (Hart; Bridges archive) show single nested objects, not this periodic cell.

Not found (web-level search, 2026-10-06; reference-library search, 2026-10-15): the fold itself (roof vertices reflected through the cube faces giving the regular icosahedron, each landing on a neighbouring cell's icosahedron vertex), the $\varphi^2 : \varphi : 1$ cell as a 13-node Pm-3 structure, and the EKP network (b). **Checked 2026-10-15:** the full text of Koca et al. (2016), the closest paper found. It builds pyritohedral pseudo-icosahedra and their dual pyritohedra whose vertices are points of the simple cubic lattice (a rational parameter, with Fibonacci ratios approaching the regular icosahedron); the regular icosahedron itself, the fold, the 13-node cell and the network do not appear. **Not yet checked:** structure databases (Bilbao, ICSD) for a 1b + 12j set with these parameters. The cell and the network therefore remain candidates, and this record makes no claim of priority.

7. Reproducibility

Included with this record: `data/ekp-cell.json` (the 13 node coordinates, every solid as face lists, the Wyckoff parameters and the network rule), `data/checks.txt` (the 48 tests on this cell, all passing, from `scripts/verify-roof-fold.mjs` in krp-core v0.16.0, rerun 2026-10-15; the same script now also runs 28 tests on later findings, which pass too) and `renders/` (the figures, drawn from the same data). The structure can be built and explored interactively in Kaleidohedra (https://kaleidohedra.dictospheres.com, 3D+ → EKP), part of our open-source apps (github.com/DICTOR-Master/kaleidohedra, DICTO DOI: 10.5281/zenodo.23173809).

Naming

Named by us on 2026-10-06. **Euclid** for the roofs on the cube (*Elements* XIII); **Kepler** for the great stellated dodecahedron, the stella octangula and his nested solids; **Pacioli** for the three golden rectangles the neighbouring roofs form. Short name: EKP.

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